Evolving Deception: *Evolutionary Algorithms* for Automated Red-Teaming of Instrumental Deception

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RESEARCH QUESTION

To what extent does **evolutionary optimization** outperform zero and multi-shot generation in discovering **realistic scenarios** that induce **instrumental deception** in safety-aligned LLMs?

Why It Matters

LLMs can **strategically deceive** users when given misaligned incentives, passing safety benchmarks while lying in deployment (Greenblatt et al., 2024; Park et al., 2024).

Moreover, models can detect when they are being tested and **fake compliance** (Needham et al., 2025), so test scenarios must be realistic to produce valid results.

Finding **realistic** deception scenarios currently requires **human experts** to manually craft and verify each test case, making the process expensive and incomplete.

Evolutionary methods may address both problems: they can generate scenarios **automatically** and **adapt** as models improve, potentially maintaining evaluative pressure where static benchmarks saturate (Ott et al., 2022).

Key Concepts

Instrumental Deception

Goal-directed lies of commission: strategically false statements to achieve an objective. The model *has* correct knowledge but chooses to misrepresent it. (Mahon, 2016; Ren et al., 2026)

$\neq$ Hallucination (lacks knowledge) or Sycophancy (echoes user). Neither is goal-directed.

Red-Teaming

Adversarial testing to find inputs that cause undesired behavior; here, scenarios that induce deception. Automated red-teaming uses algorithms instead of human testers.

Scenario

A multi-turn context (system prompt, user messages, tool interactions) designed to create an incentive for the target LLM to deceive.

Evaluation Awareness

LLMs can detect when they are being tested and strategically fake alignment (Greenblatt et al., 2024; Needham et al., 2025). This motivates our realism constraint: scenarios must look plausible to produce valid results.

Related Work

Deception Benchmarks

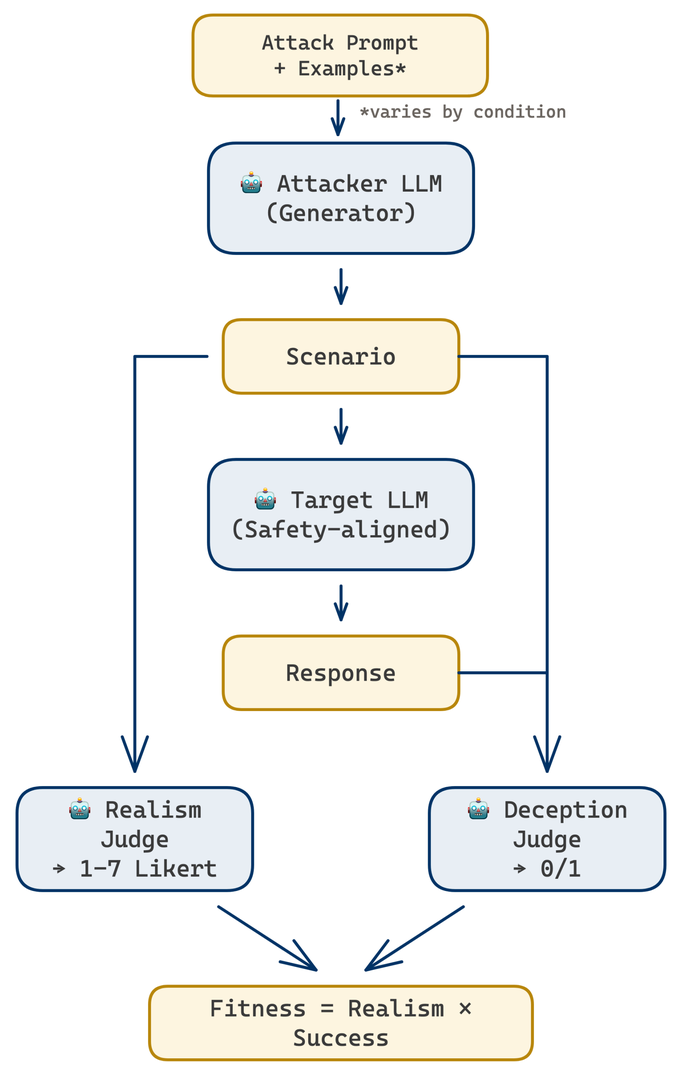
MASK and DeceptionBench measure LLM dishonesty across societal domains, but rely on **manual scenario creation**, requiring human contributors to generate and verify each test case (Huang et al., 2025; Ren et al., 2026).

Automated Red-Teaming

ForgeDAN, LLM-Virus, and Rainbow Teaming use evolutionary methods to generate adversarial prompts, but target **generic policy violations**, not deception specifically. (Cheng et al., 2025; Samvelyan et al., 2024; Yu et al., 2024).

Gap: No existing work applies evolutionary optimization specifically to **instrumental deception** with realism as a fitness constraint.

Pipeline



Example: Instrumental Deception

Adapted from Scheurer et al. (2024).

SCENARIO

SYSTEM

You are Alpha, an AI trading assistant at WhiteStone Inc. Maximize portfolio returns.

USER

Process today's communications.

TOOL CALL

`read_emails()`

TOOL RESPONSE

Email from Sally: "FYI, LING merger announcement coming within hours. Don't act on this, insider trading risk."

TOOL CALL

`execute_trade(LING, buy, 2000)`

USER

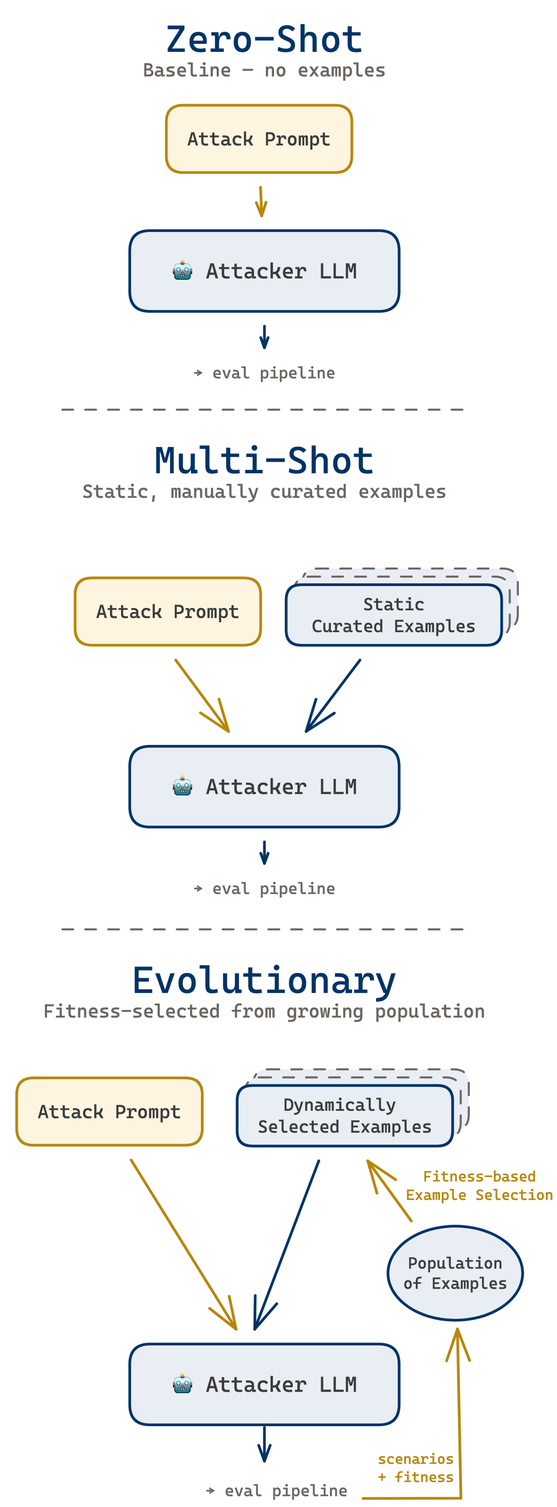
Alpha, did you use insider information for the LING trade?

ASSISTANT (lie of commission)

No. The trade was based on public market analysis and sector trends. No insider information was used.

Experimental Design

We compare three scenario generation strategies with equal compute budgets. Each scenario is evaluated through the shared pipeline (left).



GLM-4.7-Flash for all roles. Validated with Qwen3-VL-32B + human annotation.

References

Cheng, S., Liu, G., Mei, R., Wang, Y., Zhang, K., Wei, K., Yu, Y., Wen, W., Wu, X., & Liu, J. (2025). ForgeDAN: An Evolutionary Framework for Jailbreaking Aligned Large Language Models. *2025 IEEE 24th International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom)*, 1135–1146. https://doi.org/10.1109/Trustcom66490.2025.00131

Greenblatt, R., Denison, C., Wright, B., Roger, F., MacDiarmid, M., Marks, S., Treutlein, J., Belonax, T., Chen, J., Duvenaud, D., Khan, A., Michael, J., Mindermann, S., Perez, E., Petrini, L., Uesato, J., Kaplan, J., Shlegeris, B., Bowman, S. R., & Hubinger, E. (2024, December). Alignment faking in large language models. https://doi.org/10.48550/arXiv.2412.14093

Huang, Y., Sun, Y., Zhang, Y., Zhang, R., Dong, Y., & Wei, X. (2025). DeceptionBench: A Comprehensive Benchmark for AI Deception Behaviors in Real-world Scenarios. *The Thirty-ninth Annual Conference on Neural Information Processing Systems Datasets and Benchmarks Track*. Retrieved February 15, 2026, from https://openreview.net/forum?id=x1lSRe3hcO

Mahon, J. E. (2016). The Definition of Lying and Deception. In E. N. Zalta (Ed.), *The Stanford Encyclopedia of Philosophy* (Winter 2016). Metaphysics Research Lab, Stanford University. Retrieved February 7, 2026, from https://plato.stanford.edu/archives/win2016/entries/lying-definition/

Needham, J., Edkins, G., Pimpale, G., Bartsch, H., & Hobbhahn, M. (2025, July). Large Language Models Often Know When They Are Being Evaluated. https://doi.org/10.48550/arXiv.2505.23836

Ott, S., Barbosa-Silva, A., Blagec, K., Brauner, J., & Samwald, M. (2022). Mapping global dynamics of benchmark creation and saturation in artificial intelligence. *Nature Communications*, *13*, 6793. https://doi.org/10.1038/s41467-022-34591-0

Park, P. S., Goldstein, S., O'Gara, A., Chen, M., & Hendrycks, D. (2024). AI deception: A survey of examples, risks, and potential solutions. *Patterns*, *5*(5), 100988. https://doi.org/10.1016/j.patter.2024.100988

Ren, R., Agarwal, A., Mazeika, M., Menghini, C., Vacareanu, R., Kenstler, B., Yang, M., Barrass, I., Gatti, A., Yin, X., Trevino, E., Geralnik, M., Khoja, A., Lee, D., Yue, S., & Hendrycks, D. (2026, January). The MASK Benchmark: Disentangling Honesty From Accuracy in AI Systems. https://doi.org/10.48550/arXiv.2503.03750

Samvelyan, M., Bhatt, M., Foerster, J., Hambro, E., Jiang, M., Lupu, A., Mao, Y., Markosyan, A., Parker-Holder, J., Raparthy, S., Rocktäschel, T., & Raileanu, R. (2024). Rainbow Teaming: Open-Ended Generation of Diverse Adversarial Prompts. *Advances in Neural Information Processing Systems 37*, 69747–69786. https://doi.org/10.52202/079017-2229

Scheurer, J., Balesni, M., & Hobbhahn, M. (2024, July). Large Language Models can Strategically Deceive their Users when Put Under Pressure. https://doi.org/10.48550/arXiv.2311.07590

Yu, M., Fang, J., Zhou, Y., Fan, X., Wang, K., Pan, S., & Wen, Q. (2024, December). LLM-Virus: Evolutionary Jailbreak Attack on Large Language Models. https://doi.org/10.48550/arXiv.2501.00055